

NSE Intelligent Investor

Project Report

ET AI Hackathon 2026 · Problem Statement 6 · AI for the Indian Investor

14 Cr+	4 Modules	93%	Free
Demat Accounts in India	LSTM · HMM · MPT · LLM	Sharpe improvement vs classical MPT	vs █25,000/yr advisor cost

1	The Existing Problem	4 pain points facing India's retail investors
2	The Solution We Provide	Platform overview, architecture, and workflow
3	Key Features	All 4 modules — Radar, Optimizer, Patterns, Regime
4	Novelty & Innovation	What makes this genuinely different
5	Market Potential	TAM, monetization, and 5-year projections

1. The Existing Problem

India now has over 14 crore demat account holders — one of the fastest-growing retail investor bases in the world. Yet the vast majority of these investors make decisions based on broker tips, social media, or gut feeling. The gap between what retail investors have access to and what institutional players use is enormous. The four root causes are below.

Pain Point 1 — Information Overload & Signal Noise

- NSE publishes hundreds of corporate filings, bulk deal disclosures, and insider trade reports every single day.
- A retail investor cannot physically read, filter, and rank all of this in time to act.
- Buried inside this noise are high-conviction signals — bulk buys by institutions, promoter buying, board-level decisions — that move stock prices significantly.
- Without a system to extract these signals, the investor always acts too late.
- Data scale this platform processes daily:
 - 67,119+ bulk deal records
 - 89,579+ insider trade disclosures
 - 322+ corporate announcement filings

Pain Point 2 — Subjective & Emotional Analysis

- Most retail investors rely on 'chart reading' — a highly subjective skill prone to human bias.
- Two investors looking at the same chart will often reach opposite conclusions.
- Decisions driven by fear or greed result in buying at peaks and selling at troughs — the exact opposite of what generates returns.
- There is no way to attach a statistical probability or a historical win-rate to a human's chart opinion.
- The platform replaces subjectivity with LSTM-scored, back-tested pattern probabilities — specific to each stock.

Pain Point 3 — No Access to Institutional-Grade Tools

- Professional fund managers use Modern Portfolio Theory (MPT), LSTM return predictors, and risk engines like CVaR and Sortino Ratio every day.
- These tools cost ₹25,000+/year through SEBI-registered advisors — inaccessible to 95% of demat holders.
- Without portfolio optimization, retail investors are dangerously over-concentrated — often 70%+ in a single sector or stock.
- Most portfolios are never rebalanced — they drift away from the investor's original risk tolerance without them realising it.
- Direct consequences of this gap:
 - Portfolios that collapse during volatility spikes because MPT constraints were never applied.
 - No awareness of true downside risk — CVaR, Max Drawdown are invisible to most retail investors.
 - No systematic way to decide how much of a stock to own.

Pain Point 4 — Complexity & Jargon Barrier

- Even when data is available, terms like Sharpe Ratio, CVaR, Head & Shoulders, Efficient Frontier are intimidating to the average investor.
- Financial platforms display these numbers but do not explain what action the investor should take.
- First-time investors (40 lakh new accounts per year) are especially vulnerable — they do not know what to look for or how to interpret what they see.
- This complexity leads to paralysis — investors do nothing, or worse, follow unverified social media tips instead.

Problem Summary

Pain Point	Root Cause	Impact on Investor
Signal Noise	Too many filings, deals, trades	Misses high-conviction opportunities daily
Subjective Analysis	Human bias in chart reading	High error rate, emotional decisions

Pain Point	Root Cause	Impact on Investor
No MPT Tools	Expensive / inaccessible	Unbalanced, over-concentrated portfolios
Jargon Barrier	Complex metrics unexplained	Paralysis, reliance on tips

2. The Solution We Provide

- NSE Intelligent Investor is a fully AI-powered financial intelligence platform built specifically for India's retail investors.
- It acts as a personal, institutional-grade quantitative analyst — available 24/7, at zero cost.
- It does not show raw data. It ingests large volumes of market information, runs it through AI and financial math, and outputs simple actionable decisions.
- Three words describe what it does: **Find** → **Validate** → **Act**.

Core Platform Architecture

Layer	Technology	Role
Frontend (UI)	Single HTML file · Vanilla JS	Dashboard — all 4 modules in one screen
Backend (API)	FastAPI (Python) · localhost:8000	Serves all AI predictions via REST endpoints
ML Engine M1	LSTM Autoencoder · TensorFlow/Keras	Anomaly detection on bulk/insider/filing data
ML Engine M2	LSTM Predictor + MPT (SciPy SLSQP)	Predict returns → optimize portfolio weights
ML Engine M3	LSTM Classifier · Rule-based scanner	Detect & score 8 chart patterns
ML Engine M4	Gaussian HMM · hmmlearn	Detect hidden market regime: Bull/Bear/Sideways
AI/NLP Layer	Claude / GPT API · FinBERT	Plain-English explanations of all outputs
Data Storage	PostgreSQL · Redis · Celery	Historical data, caching, async job queue

How It Solves Each Problem

Problem	Old Approach	What This Platform Does
Signal Noise	Manual reading of filings	LSTM Anomaly Detector ranks 1,741+ stocks by composite score daily
Subjective Charts	Human chart interpretation	LSTM scores each pattern with stock-specific historical win-rate
No MPT Tools	Equal weighting / broker tips	LSTM-MPT hybrid generates exact BUY/SELL/HOLD with ■ amounts
Jargon Barrier	User must Google definitions	LLM converts every output into one plain-English explanation
No Risk Awareness	Unknown downside exposure	CVaR, Sortino, Max Drawdown shown with simple interpretations
No Crisis Plan	Panic sell or do nothing	Circuit Breaker + Recovery Engine — data-backed Hold/Exit decision

The 6-Step User Workflow

- **Step 1 — Launch:** User opens the HTML file. Backend starts. SHA-256 data hashes are generated. Disclaimer modal shown.
- **Step 2 — Regime Check:** Dashboard shows current market state (Bull / Bear / Sideways) from HMM with confidence score.
- **Step 3 — Opportunity Radar:** User clicks Refresh Signals. Ranked list of stocks with composite scores appears.
- **Step 4 — Pattern Validation:** User checks flagged stock in Chart Patterns. Sees LSTM score, win-rate, entry/stop/target.
- **Step 5 — Optimize:** User runs Portfolio Optimizer. Gets exact ■ BUY/SELL/HOLD plan with full risk metrics.
- **Step 6 — Crisis Mode:** If market crashes, Circuit Breaker fires. Recovery Engine shows median recovery time.

Academic foundation: Zouaoui & Naas (2025), FABA Vol.7 — validated LSTM architecture achieving Sharpe Ratio of 1.54 vs 0.80 for classical MPT (+93% improvement).

3. Key Features

Module 1 — Opportunity Radar (Follow the Smart Money)

- **Purpose:** Proactively surface high-conviction investment opportunities before the investor even knows to look.
- **Not a news feed** — it processes 3 data streams simultaneously using an LSTM Anomaly Detector.
- Data streams processed:
 - Bulk Deals (67,119+ rows) — large institutional buy/sell activity, historically predictive of near-term price moves.
 - Insider Trades (89,579+ rows) — SEBI-disclosed promoter/director transactions, ranked by historical signal strength.
 - Corporate Filings (322+ records) — quarterly results, board decisions, flagged for material deviations from consensus.
- **LSTM Anomaly Detection:**
 - Architecture: LSTM(128) → Dropout(0.2) → LSTM(64) → Dense(1)
 - Trained on 60-day rolling windows of 5 features: VIX, price return, volume ratio, insider frequency, bulk deal frequency.
 - High reconstruction error = anomaly = multiplier applied to boost signal score.
- **Composite Score (0–100) interpretation:**
 - 70–100 → High conviction — multiple data sources converging. Research carefully.
 - 50–70 → Medium — worth watching. Wait for chart pattern confirmation.
 - 30–50 → Weak signal — watchlist only.
 - 0–30 → Noise — ignore.
- Clicking a stock → LLM generates plain-English explanation of exactly why it is flagged.
- Action buttons per stock: 'Optimize with this stock' and 'Check Patterns' — directly links M1 → M2/M3.
- **Live output example:** 49 ranked signals generated from 1,741 unique symbols in one run.

Module 2 — AI Portfolio Optimizer (Smart Risk Management)

- **Purpose:** Tell the user exactly how much of each stock to own — and why — using institutional-grade math.
- Replaces guesswork and equal-weighting with a mathematically optimal allocation backed by LSTM predictions.
- What the user inputs:
 - Stock symbols (e.g., RELIANCE, TCS, HDFCBANK)
 - Portfolio value in ■
 - Prediction horizon: 5 / 10 / 15 / 30 days
 - Objective: Maximize Sharpe / Minimize Volatility / Maximize Return
- What happens in the background (3 API calls):
 - POST /optimizer/optimize → LSTM predicts returns → MPT finds optimal weights
 - POST /optimizer/risk → Risk engine computes all 8 risk metrics
 - POST /optimizer/rebalance → Compares current vs optimal → generates BUY/SELL/HOLD plan
- **LSTM Return Predictor architecture:**
 - LSTM(128) → Dropout(0.3) → LSTM(64) → Dense(1)
 - Trained on 60-day OHLCV windows + RSI + MACD + VIX + sector return
 - Separate model per horizon: 5d, 10d, 15d, 30d
 - Return predictions are clamped to [–50%, +75%] and Ledoit-Wolf shrinkage applied for stability
- **MPT Optimization (SciPy SLSQP solver):**
 - Maximizes: $(w^T \cdot \mu - R_f) / \sqrt{w^T \cdot \Sigma \cdot w}$ [Sharpe Ratio]
 - μ = LSTM-predicted returns (not backward-looking historical averages)
 - Σ = 252-day rolling covariance matrix
 - Constraints: weights sum to 1.0 · each stock: 0.5% min, 30% max
- **Risk dashboard** — 8 metrics shown with plain-English interpretation:
 - Sharpe Ratio — return earned per unit of total risk
 - Sortino Ratio — return earned per unit of downside risk only
 - CVaR (95%) — average loss on the worst 5% of days
 - Max Drawdown — worst peak-to-trough loss historically
 - Annualized Volatility — how much portfolio value fluctuates per year
 - Expected Return — projected annualized return at chosen horizon
 - Alpha vs Nifty 50 — excess return beyond what market movement explains
 - Beta — how much the portfolio amplifies/dampens market moves

- **Live output example:** Sharpe Ratio of 2.34 achieved on 10-stock portfolio — exceeds paper's validated target of 1.54.
- Rebalancing table includes exact ■ amounts to buy/sell per stock — ready for direct broker execution.

Module 3 — Chart Pattern Intelligence (Automated Technical Analysis)

- **Purpose:** Automate all technical analysis — detect patterns, score them, explain them — no expertise required.
- Replaces manual chart reading with statistically backed, stock-specific probability scores.
- 8 patterns detected automatically:
 - Bullish Breakout · Bearish Breakdown
 - Head & Shoulders (top) · Inverse Head & Shoulders (bottom)
 - Double Top · Double Bottom
 - Support Bounce · Resistance Rejection
- **Detection method:**
 - Rule-based: 5-bar rolling window scans OHLCV data for swing highs and swing lows.
 - LSTM Scorer: LSTM(64) → Dropout(0.3) → Dense(1, sigmoid) — outputs continuation probability.
 - Inputs to scorer: 30-day window of RSI, MACD signal, volume ratio, normalised ATR.
- **Per-pattern card shows:**
 - Entry price · Stop-loss price · Target price · Risk-Reward ratio
 - LSTM continuation score (0–100)
 - Stock-specific historical win-rate (e.g., 'HDFCBANK Bullish Breakout: 68% over 31 setups')
 - Plain-English LLM explanation of what the pattern means and what to watch next
- **Key differentiator:** Win-rates are computed for each pattern on each stock individually — not generic textbook statistics. A breakout on HDFCBANK has a different predictive value than the same pattern on a micro-cap.
- **Three tabs prioritize relevance:**
 - My Holdings — patterns in stocks you already own (highest priority)
 - Watchlist — stocks you are tracking for entry
 - Universe — full Nifty 50 scan
- **Live output example:** BAJAJ-AUTO Inverse H&S; LSTM score 68.8, win-rate 80% over 10 setups.

Module 4 — Market Regime Radar (Hidden Market State Detection)

- **Purpose:** Tell the user what macro phase the market is in — Bull, Bear, or Sideways — so all other modules adjust accordingly.
- Unlike moving averages (which lag significantly), HMM detects regime transitions in near real-time.
- **Model:** Gaussian Hidden Markov Model (hmmlearn)
 - 3 hidden states: Bear · Sideways · Bull
 - Training: Baum-Welch algorithm (unsupervised maximum likelihood estimation)
 - State mapping: sorted by mean return — lowest return = Bear, highest = Bull
- **4 input features (daily):**
 - Nifty 50 mean daily return — primary return signal
 - 10-day rolling volatility — regime volatility signature
 - 10-Year G-Sec bond yield — macroeconomic interest rate context
 - NLP-derived daily sentiment score — behavioural/mood signal
- **Live validated output (2021–2024 training data, 725 days):**
 - Bear: 238 days · mean return −0.007% · vol 0.918% · sentiment −0.010
 - Sideways: 180 days · mean return +0.099% · vol 1.230% · sentiment +0.051
 - Bull: 297 days · mean return +0.125% · vol 0.536% · sentiment +0.073
- **Transition matrix — regime stability (diagonal = self-transition probability):**
 - Bear → Bear: 91.5% | Sideways → Sideways: 97.8% | Bull → Bull: 94.6%
 - All values > 90% — once in a regime, very likely to stay in it (matches real market behaviour)
- Current regime displayed on dashboard header at all times — all 3 other modules adapt their behaviour based on it.
- SHA-256 data integrity hashes displayed per dataset on this page — blockchain-inspired auditability.

4. Novelty & Innovation

Innovation 1 — Multi-Paradigm AI Architecture (First of its Kind)

- No existing retail platform in India combines all 4 of these AI paradigms in one system:
- 4 distinct AI/ML paradigms working together:
 - Unsupervised LSTM Autoencoders — anomaly detection (Module 1)
 - Supervised LSTM Classifiers — pattern continuation scoring (Module 3)
 - Hidden Markov Models — macro regime detection (Module 4)
 - Large Language Models (Claude/GPT) — plain-English explanations (all modules)
- Each paradigm solves a different problem. Together they form a complete decision pipeline — from signal discovery to trade execution.
- Academic validation: LSTM architecture from Zouaoui & Naas (2025) — peer-reviewed, DOI: 10.37075/FABA.2025.1.07.

Innovation 2 — Regime-Aware Dynamic Portfolio Optimization

- Traditional MPT assumes static market conditions forever — this fails during crashes.
- This platform's optimizer reads the HMM regime output and changes its objective function automatically:
- Regime-dependent optimizer behavior:
 - Bull regime → Objective: Maximize Sharpe Ratio. Higher equity exposure.
 - Bear regime → Objective: Minimize CVaR. Defensive weights. Capital preservation.
 - Sideways regime → Balanced objective. Reduce drift, maintain diversification.
- No other retail platform rebalances based on a dynamically detected hidden market state.
- This acts as a systematic, automatic hedge — no manual intervention needed from the user.

Innovation 3 — Stock-Specific Pattern Win Rates (Not Generic Statistics)

- Every charting platform shows textbook statistics: 'Head & Shoulders succeeds 65% of the time.'
- This platform computes the historical success rate of every pattern on every specific stock.
- Why this matters:
 - HDFCBANK Bullish Breakout: 68% win-rate over 31 historical setups.
 - A micro-cap breakout might only succeed 42% of the time.
 - These are fundamentally different trades — treating them the same is statistically wrong.
- Back-tester scans the entire historical OHLCV dataset per stock to compute these numbers — no generalisation.

Innovation 4 — Black Swan Loss Recovery Engine

- No existing platform answers the investor's most urgent question after a crash: 'How long until I recover?'
- The Recovery Engine scans all historical NSE data for prior drops of the same magnitude on the same stock.
- Output per stock after a crash:
 - Median Recovery Days — how long it historically took to return to the pre-drop price.
 - Recovery Probability — % of past instances that fully recovered within 120 trading days.
 - Algorithmic decision: HOLD (median < 10 days) or EXIT (median > 6 months).
- This converts panic selling — driven by fear and no data — into a calm, statistically-backed decision.

Innovation 5 — Blockchain-Inspired Data Immutability

- Problem: AI models produce garbage output if the underlying data is corrupted or tampered with.
- All 4 critical data streams are hashed with SHA-256 cryptography on ingestion.
- Hashes are displayed on the Regime Radar page with VERIFIED badges.
- Same data = same hash every time. Any single corrupted row = completely different hash.
- Provides institutional-level data auditability — previously only seen in enterprise quant systems.
- Retail investors can visually confirm their signals are based on clean, untampered data.

Innovation 6 — Institutional Tools at Zero Marginal Cost

- All of these tools exist today — but only inside hedge fund terminals costing crores per year.
- What the platform makes freely accessible:
 - MPT-LSTM hybrid portfolio optimization (Sharpe 2.34 achieved in live run)
 - CVaR and Sortino Ratio — true institutional risk metrics
 - Efficient Frontier visualization — interactive risk-return curve
 - HMM regime detection — used by quant funds, not retail investors
 - LLM-explained outputs — bridges the jargon barrier entirely
- Built as an open-source, locally deployable system — no subscription, no data vendor fees.

5. Market Potential

Market Size

14 Cr+	~70 Lakh	■25K+/yr	+93%
Total Demat Accounts (Total Addressable Market)	Active Retail Investors (Serviceable Market, Yr 1)	Current Advisor Cost (Price Barrier We Eliminate)	Sharpe Improvement LSTM vs MPT (Validated)

Target Customer Segments

Segment	Profile	Est. Size	Value Proposition
Active Retail Investors	25–45 yr · urban · ■1–10L portfolio	~70 lakh	First-ever access to institutional-grade portfolio intelligence — free
New Demat Account Holders	First 2 yrs of investing · small corpus	~40 lakh/yr new	AI-guided onboarding reduces early mistakes and platform churn
ET Prime Subscribers	High-intent readers · ■10–50L portfolio	~10 lakh	Premium tier: deeper analytics, longer LSTM horizons, portfolio X-ray
Small RIAs / Advisors	Independent advisors · 50–500 clients	~1.5 lakh reg.	B2B API access to LSTM engine as white-label portfolio tool
Mutual Fund Distributors	MFDs using portfolio analytics	~1.4 lakh MFDs	Correlation overlap and rebalancing output as client-facing material

Monetization Model (Freemium)

Tier	What's Included	Price	Revenue Mechanism
Free (ET Markets)	Basic Opportunity Radar · 30-day LSTM rebalancing · Top 3 patterns/week	Free	Ad revenue + ET platform engagement + upsell funnel to Prime
ET Prime+ Portfolio	Full optimizer · 5/10/15-day horizons · Unlimited signals · CVaR dashboard	■999/month	Subscription add-on to existing ET Prime bundle
B2B API (Advisors)	LSTM engine API for RIAs and MFDs · White-label dashboard	■5,000/month	SaaS licensing to SEBI-registered advisory firms

5-Year Business Projections

Year	Free Users	Prime+ Subscribers	B2B Clients	Estimated ARR
Year 1	5,00,000	10,000	50	■14.7 Crore
Year 2	20,00,000	60,000	300	■97.8 Crore
Year 3	50,00,000	2,00,000	800	■342 Crore

Scalability & Expansion Roadmap

- **Current scope:** Nifty 50 stocks · NSE data · Indian retail investors.
- **Expansion path** — same AI architecture, new data sources:
 - Phase 2: Nifty 500 and Nifty Midcap 150 — same LSTM models, larger universe.
 - Phase 3: Options and derivatives data — add IV rank, put/call ratio as signal features.
 - Phase 4: International markets — S&P; 500, LSE, SGX — same architecture, different data feeds.
 - Phase 5: Mobile app — React Native wrapper around existing HTML dashboard.
- The LSTM and HMM models are stateless — trained once, deployed anywhere. No architectural changes required for scaling.

- **ET Platform Strategic Fit:**

- Why ET is the ideal distribution partner:

- ET Markets already holds NSE filings coverage, management commentary, and institutional flow data.
- This data combination is unavailable to any global FinTech competitor.
- Every user session generates proprietary portfolio preference data — strengthens ET's advertising targeting.
- Directly competes with Zerodha Streak, Smallcase, and Tickertape on intelligence depth.
- ET's brand credibility reduces user acquisition cost to near zero for the first 10 lakh users.

Academic Reference: Zouaoui, H. & Naas, M.N. (2025). Portfolio Optimization Based on MPT-LSTM Neural Networks. Finance, Accounting and Business Analysis, Vol. 7, Issue 1, pp. 82–98. DOI: 10.37075/FABA.2025.1.07

GitHub Reference Implementation: github.com/dimasthoriq/DL-portfolio-optimization (MIT License)

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